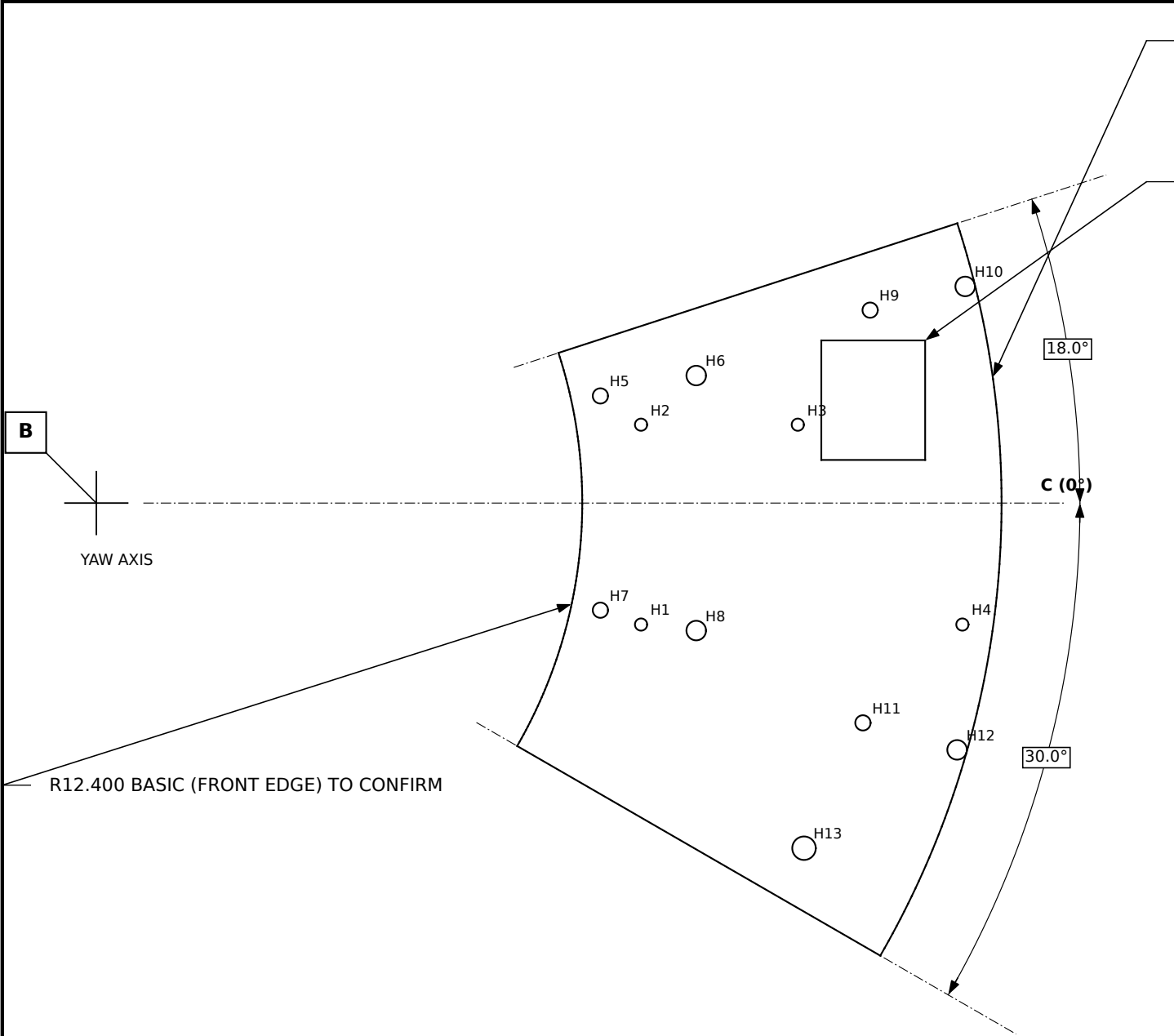




R23.100 BASIC

OUTLINE

	.020	A	B	C
--	------	---	---	---

NOTCH THRU: X 18.50-21.15, Y 1.10-4.15 BASIC
(CLEARS THE ROLL NEMA 23 AT +20° PITCH)

DATUM A: TOP FACE

	.005
--	------

HOLE TABLE (BASIC FROM B / C) - POSITION $\varnothing$.010 A B C

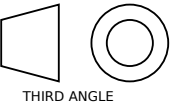
HOLE	SIZE	X	Y	R	θ	FOR
H1	3/8-16 UNC-2B THRU	13.900	-3.100	-	-	SECTOR BASE
H2	3/8-16 UNC-2B THRU	13.900	+2.000	-	-	SECTOR BASE
H3	3/8-16 UNC-2B THRU	17.900	+2.000	-	-	SECTOR BASE
H4	3/8-16 UNC-2B THRU	22.100	-3.100	-	-	SECTOR BASE
H5	$\varnothing$.390 THRU	12.863	+2.734	13.150 TO CONFIRM	+12.0°	WHEEL STUD
H6	$\varnothing$.500 THRU	15.308	+3.254	15.650 TO CONFIRM	+12.0°	ECC. SPACER
H7	$\varnothing$.390 THRU	12.863	-2.734	13.150 TO CONFIRM	-12.0°	WHEEL STUD
H8	$\varnothing$.500 THRU	15.308	-3.254	15.650 TO CONFIRM	-12.0°	ECC. SPACER
H9	$\varnothing$.390 THRU	19.746	+4.923	20.350 TO CONFIRM	+14.0°	WHEEL STUD
H10	$\varnothing$.500 THRU	22.171	+5.528	22.850 TO CONFIRM	+14.0°	ECC. SPACER
H11	$\varnothing$.390 THRU	19.562	-5.609	20.350 TO CONFIRM	-16.0°	WHEEL STUD
H12	$\varnothing$.500 THRU	21.965	-6.298	22.850 TO CONFIRM	-16.0°	ECC. SPACER
H13	$\varnothing$.600 THRU	18.059	-8.808	20.092	-26.0°	PINION SHAFT

PLAN VIEW (TOP) 1:4

NOTES:

- DIMENSIONS MARKED TO CONFIRM DEPEND ON THE DOWNSTREAM BAY FLOOR HEIGHT/WIDTH AND THE SPACE UNDER THE TEST-SECTION EXIT; VERIFY ON THE TUNNEL BEFORE MACHINING.
- DATUM A = TOP FACE (SECTOR BASE SEATS HERE), FLAT .005. DATUM B = YAW AXIS (CENTRE OF THE ARCS). DATUM C = PLAN ANGLE 0 (+X DOWNSTREAM). HOLE TABLE IN BASIC COORDINATES FROM B ALONG C.
- WHEEL STUD HOLE RADII ASSUME A 1.00 OD WHEEL TANGENT TO THE RAIL V-EDGE APEX: RESET THEM FROM THE PURCHASED WHEEL GROOVE DEPTH (TO CONFIRM). OUTER HOLE OF EACH TRUCK IS .500 FOR THE ECCENTRIC SPACER (WT-6C-506-2).
- FRONT (INNER) EDGE R12.400 PASSES UNDER THE TEST-SECTION EXIT AT +-15 DEG YAW ($x = 10.7$ IN).
- LIGHTENING POCKETS AFTER FEA (NOT SHOWN).

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
.X $\pm$.03 .XX $\pm$.010
.XXX $\pm$.005 .XXXX $\pm$.0005
ANGLES $\pm$ 0°30'
BREAK SHARP EDGES .005-.015
SURFACE FINISH 63 μ in Ra MAX
INTERPRET PER ASME Y14.5-2018
UNDIMENSIONED GEOMETRY PER STEP MODEL



NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
PLATE, YAW CARRIAGE								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE (MIC-6 PREFERRED)				CLEAR ANODIZE TYPE II				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-504		A	B	1:4		7.27 lb		34 OF 36

PRELIMINARY - TO CONFIRM